Notes on Computational Fluid Dynamics: General Principles

5.3 Convergence

The Gauss-Seidel method was demonstrated in Sec. 5.2 using a sample problem, Eq. (5.2), that converged to within 0.2% accuracy in 9 solution sweeps. This section discusses the criteria for convergence of a solution.

The easiest explanation of convergence considers the error ϵ_i for each equation i, defined in Sec. 5.2. Substituting $\Psi_i = (\Psi_i)_{\text{ex}} + \epsilon_i$ in each term in Eq. (5.3), *e.g.* in Eq. (5.3a), gives

$$(\Psi_1)_{\text{ex}} + \epsilon_1 = \frac{1}{3}\left(6 + (\Psi_2)_{\text{ex}} + \epsilon_2 + (\Psi_2)_{\text{ex}} + \epsilon_3\right). \tag{5.5}$$

Since $3(\Psi_1)_{\text{ex}} - (\Psi_2)_{\text{ex}} - (\Psi_3)_{\text{ex}} \equiv 6$, Eq. (5.5) reduces to

$$\epsilon_1 = \frac{1}{3}\epsilon_2 + \frac{1}{3}\epsilon_3. \tag{5.6}$$

The magnitude of error $|\epsilon_1|$ is *at least* as large as the sum of the terms on the r.h.s., but is smaller if the signs of ϵ_2 and ϵ_3 are different, *i.e.*

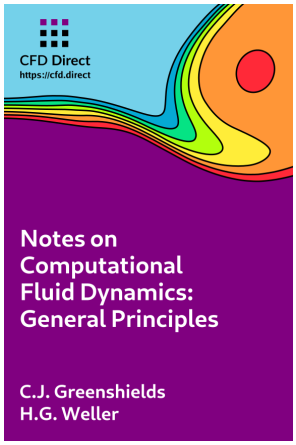
$$|\epsilon_1| \leq \frac{1}{3}|\epsilon_2| + \frac{1}{3}|\epsilon_3|. \tag{5.7}$$

Repeating for Eq. (5.3b) and Eq. (5.3c) gives:

$$|\epsilon_2| \leq \frac{1}{5}|\epsilon_1| + \frac{2}{5}|\epsilon_3|; \tag{5.8a}$$

$$|\epsilon_3| \leq \frac{1}{4}|\epsilon_1| + \frac{1}{2}|\epsilon_2|. \tag{5.8b}$$

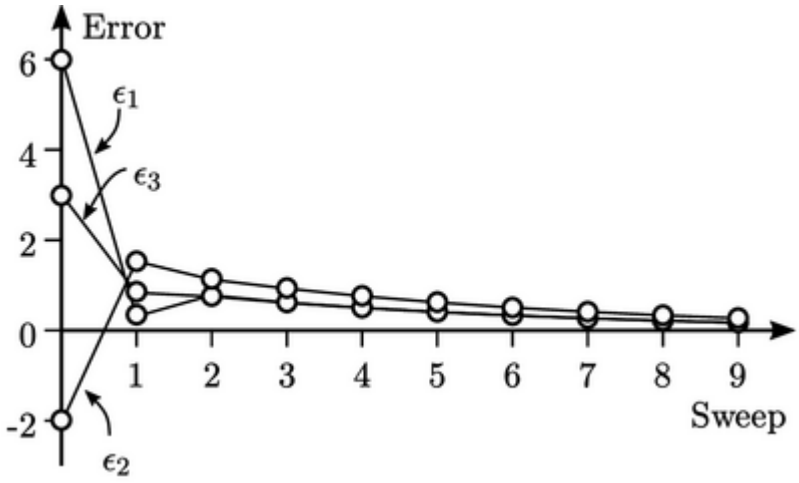
The solution begins with an initial error of $\epsilon_1 = 6$, $\epsilon_2 = -2$ and $\epsilon_3 = 3$. After one sweep the error is $\epsilon_1 = 0.33$, $\epsilon_2 = 1.53$ and $\epsilon_3 = 0.85$.



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The error is quickly distributed evenly, such that ϵ_1 and ϵ_3 are almost identical at sweep 2. The errors continue to reduce since Eq. (5.7) and Eq. (5.8) guarantee that no error is greater than the average of the other errors.

Condition for convergence

The behaviour of this problem indicates a convergence condition for the Gauss-Seidel method: the magnitude of the diagonal coefficient in each matrix row must be greater than or equal to the sum of the magnitudes of the other coefficients in the row; in one row at least, the "greater than" condition must hold.

This is known as *diagonal dominance*, which is a *sufficient* condition for convergence, described mathematically as

$$|a_{i,i}| \geq \sum_{\substack{j=1 \\ i \neq j}}^{N} |a_{i,j}| \quad \text{for all } i, \tag{5.9}$$

where the '$>$' condition must be satisfied for at least one i. The description of the condition as "sufficient" means that convergence may occur when the condition is not met.

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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